AquaTrack

Water Tanker Management System

Smarter water delivery for drought-prone districts

Full-Stack Hackathon Project | React · Node.js · MongoDB · Socket.io

The Problem

🏜️ Water Scarcity

Drought-prone districts face chronic water shortages affecting millions of rural residents every year.

📋 Manual Scheduling

Delivery planning is done manually — prone to errors, delays and inequitable distribution.

📍 No Real-Time Visibility

Village leaders have no way to track tanker locations or know when water will arrive.

📢 Complaint Backlog

Unresolved grievances pile up with no structured channel for escalation or follow-up.

Our Solution — AquaTrack

A centralized platform unifying delivery management, live tracking, and community feedback

AI Demand Scoring

Priority villages ranked by population × consumption − last delivered

Live GPS Tracking

Real-time tanker position streamed via Socket.io

Fleet Management

Full tanker lifecycle — capacity, status, driver assignment

Complaint System

Structured grievance portal with priority triage and resolution notes

Role-Based Auth

JWT-secured access for Admin, Driver and Village Leader roles

Alert Engine

Auto-flags delayed deliveries and emits real-time socket alerts

System Architecture

Frontend (React + Vite)

- Dashboard · Tracking · Deliveries
- Villages · Tankers · Complaints
- Socket.io Client · Recharts

→

Backend (Node.js + Express)

- REST API + Socket.io Server
- JWT Auth Middleware
- Demand Prediction · Alert Services

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Database (MongoDB)

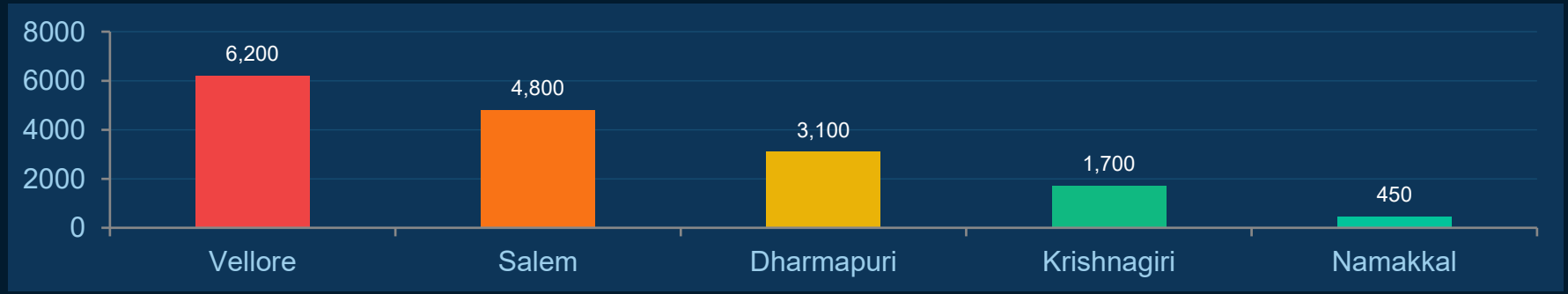
- Users · Villages · Tankers
- Deliveries · Complaints · Alerts
- DeliveryLog (audit trail)

AI Demand Prediction Engine

Demand Score = (Population × Avg. Daily Consumption) − Last Delivered Liters

Higher score = greater urgency. Shortage flagged when score > population × 5 × 2

CRITICAL	HIGH	MEDIUM	LOW
Score > 5,000	Score > 2,000	Score > 500	Score ≤ 500



Real-Time GPS Tracking via Socket.io

1. **Admin dispatches tanker**

   Delivery record created with origin & destination coords

2. **start_tracking event emitted**

   Client sends socket event with delivery ID & GPS endpoints

3. **Server interpolates 20 steps**

   setInterval fires every 1.5s, emitting delivery_location_update

4. **Live map updates**

   All connected clients receive position; map pin animates in real time

5. **delivery_completed_tracking**

   Final event fired; tanker status returns to 'available'

Data Models

User

- name, email, password
- role: Admin|Driver|VillageLeader
- village (ref), phone, isActive

Village

- name, district, population
- location {lat, lng}
- demandScore, isShortage, lastDeliveryDate

Tanker

- vehicleNumber, capacityLiters
- status: available|busy|maintenance
- driver (ref), currentLocation

Delivery

- tanker, village, driver (refs)
- status: pending|in-progress|completed
- quantityScheduled/Delivered, isDelayed

Complaint

- raisedBy, village, delivery (refs)
- subject, description, priority
- status: open|in-review|resolved

Alert

- type: delay|shortage|complaint
- severity: info|warning|critical
- isRead, village, delivery (refs)

Technology Stack

Frontend	React 18 + Vite	Socket.io Client	Recharts	Context API (Auth)
Backend	Node.js + Express	Socket.io Server	JWT (bcryptjs)	Mongoose ODM
Database	MongoDB (Atlas/Local)	Mongoose Schemas	Indexed queries	Seed script
Tooling	Vite dev server	CORS middleware	dotenv config	.code-workspace

REST API Overview

/api/auth	POST /register POST /login
/api/villages	GET / POST / GET /:id/demand PUT /:id
/api/tankers	GET / POST / PUT /:id DELETE /:id
/api/deliveries	GET / POST / PUT /:id/status GET /:id
/api/complaints	GET / POST / PUT /:id/status
/api/dashboard	GET /stats GET /alerts PUT /alerts/:id

Impact & Conclusion

3

Roles

Admin · Driver · VillageLeader

6+

Modules

Auth · Villages · Tankers · Deliveries · Complaints · Dashboard

20

Steps

Interpolated GPS waypoints per live delivery

Real

-time

Socket.io event-driven alerts & location updates

Key Takeaways

- ✓ Prioritizes water equity using an AI-style demand scoring algorithm
- ✓ Real-time visibility eliminates uncertainty for village leaders and admins
- ✓ Modular architecture makes it easy to extend (mobile app, IoT sensors, ML forecasting)
- ✓ Production-ready auth, error handling, and seed data for rapid demo deployment